Impact and Ethics Note

Diabetes risk screening in primary care · Tech Wave 2026, Track 6 · **Your Team Name: Member 1, Member 2, Member 3**

Prototype. This tool estimates risk to support a clinician. It is not a diagnosis and not a replacement for clinical judgement or medical advice. A blood test (fasting glucose or HbA1c) confirms diabetes.

WHAT IT IS AND WHO BENEFITS

A model plus a screening tool (live demo, English, Français, Kreol Morisien) that turns 16 simple questions into a LOW, MODERATE or HIGH risk band, shows the factors behind the score, and suggests a follow-up date. **Who benefits:** adults who would otherwise be diagnosed late (IDF 2024: 20.1% of Mauritian adults have diabetes and 28.2% of them are undiagnosed); nurses and GPs who need a fast, explainable second opinion; and health planners who can target follow-up.

DATA, PRIVACY AND RESULTS

Data: public CDC BRFSS 2015 survey, 253,680 responses, 13.9% prevalence. No real patient data is used or collected. Income, education and US-insurance variables are not model inputs. **Privacy:** no accounts, no storage, no inputs sent to any server; users are told not to enter names or IDs. **Results (held-out test set):** ROC-AUC 0.82 (gradient-boosting benchmark 0.83); at the screening threshold sensitivity 80%, specificity 69%, NPV 95%, PPV 29%.

RISKS AND HOW WE REDUCE THEM

Read as a diagnosis, or over-trusted	Prototype banner in all three languages; wording says "estimated risk"; blood-test confirmation advised; clinician decides.
False reassurance	At the screening threshold the tool misses about 20% of true cases, so even a LOW band suggests re-screening in 3 years. We state this openly.
Unequal performance by age	Only 38% of true cases aged 18-39 are flagged, versus 87% at 70+, because risk is low in the young. Clinicians should not take a LOW result as reassurance for a younger adult with strong risk factors.
Unequal burden by income and education	Lower-income adults are flagged more often (false-positive rate 49% vs 22% for higher income), which means more confirmatory tests for them. Neither variable is a model input; both are audited.
US data, not Mauritian	Probabilities are labelled as not calibrated for Mauritius. The data has no ethnicity variable, so fairness across Mauritian communities is untested and must be checked before any pilot.
Translation errors	French and Kreol Morisien text to be reviewed by native speakers and a clinician before real use.

HOW IT COULD BE DEPLOYED

1. Now: open-source prototype and clinician feedback. **2. Validate:** recalibrate on de-identified Mauritian data with ethics approval and the Ministry of Health, including community-level fairness and age-specific checks. **3. Pilot:** supervised use in a few primary-care centres, measuring cases found, follow-ups completed and referral burden. **4. Maintain:** regular re-audit of accuracy and fairness, with a named clinical owner and a way for clinicians to report errors.

Sources: IDF Diabetes Atlas, Mauritius (diabetesatlas.org/data-by-location/country/mauritius); CDC BRFSS 2015 (Kaggle mirror: Diabetes Health Indicators Dataset). Code: see repository README.